Ziyao Tian

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AI Safety Researcher with 3+ years shipping production machine learning systems.

AI SAFETY PROGRAMS & COMMUNITY

Anthropic Fellows Program (AI Safety) — advanced to final interview — Aug 2026

Singapore AI Safety Hub Fellowship — waitlisted after final interview — Aug 2026

BlueDot Impact Technical AI Safety Course — Sep 2025 - Dec 2025

Trajectory Labs - Canada's Biggest AI Safety Community — May 2025 - Present

- Led weekly discussions, reading groups and hackathons on AI safety topics

AI Red Teamer work test — Oct 2026

- Successful indirect prompt injection against a frontier ai agent and direct jailbreak prompts against Opus 5

AI SAFETY RESEARCH

Long-Term Adversarial Memory Safety (LAMS) — Nov 2025 - Present

- Ran a controlled pilot (Gemma-2-2B, N=60 paired queries) testing whether a prior jailbreak turn contaminates a later benign turn — a deployment-time attack surface that weight-level refusal training doesn't touch
- Found post-adversarial answers diverge ~5× from baseline (0.119 vs 0.022 cosine distance, 8/60 derailments)

Memory Utilization Gap — May 2026 - Present

- Found **request-gated personalization**: models adapt to a user's cultural context **~70%** of the time when asked directly, but **<1%** spontaneously when that same context sits in long-term memory — and explicit instructions to personalize fail to move it
- Key measurement insight: query-cued evaluation cannot observe unprompted memory use, because the query is itself the trigger — standard memory benchmarks are structurally blind to this failure mode
- Designed a 6-condition behavioral evaluation of user-memory personalization; ran **1,300+ generations** across Gemini and Gemma via provider APIs with elicitation controls, quality auditing, and a reproducible pipeline
- Validated scoring with LLM-as-judge plus human annotation (annotation codebook, judge error auditing); documented stereotyping in adapted outputs as a distinct failure mode

Model Organisms: creating model organisms of misalignment — Jun 2025

- Designed a 150-example benchmark isolating persuasion-over-accuracy failure, then LoRA fine-tuned Qwen-0.5B to test how benchmark design affects post-training outcomes
- Achieved **10×** higher cross-domain generalization than a 7,000+ example baseline — demonstrating that targeted evaluation design outperforms data volume in eliciting the target behavior
- Automated synthetic data generation with tone variation and temperature sweeps for reproducibility across runs

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slate Technologies — Jul 2022 - Nov 2023

- Developed **semantic search** (+55% recall / +15% precision) by customizing a cross-encoder on proprietary data
- Implemented a constraint-aware **RL scheduler** to optimize project durations
- Built an **auto-tagging** system achieving 95% recall / 98% precision

Data Scientist | Surgical Safety Technologies — Oct 2020 - Jun 2022

- Improved ResNet latency **4x** and recall **+22%** for intraoperative image classification in a safety-critical clinical setting
- **Tripled** average weekly users by developing a recommendation algorithm
- Designed and implemented KPIs from HL7 data

Research Assistant | University of Waterloo (advisor: Prof. Samuel Wong) — May 2019 - Sep 2019

- Generated piano sonatas in the style of Beethoven using LSTM

LLM TOOLING

Clean Chat — Jul 2026 - Present

- Built an open-source tool to transform linear chat threads into dynamic tree-structured learning interface across 19 LLM sites and Claude Code, aligning with how humans read, explore and learn
- Implemented features to collapse answers you understood, highlight text, ask follow ups on subquestions — one click to make it simpler or give an example, and suggest questions that fill in your knowledge gaps

EDUCATION

University of Waterloo — Bachelor of Mathematics, Honours in Statistics — 2016 - 2020

Millsaps College — Philosophy (ethics and decision-making), transferred — 2014 - 2016